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BSIT Capstone Project Proposal Outline

Title of the Project	Implementation of Web-Based Multi-Business, Monitoring with Inventory and Financial Reporting for Office of Business Affairs
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Adviser	Mr. Jayson Galang
Rationale	The development of the Web-based Integrated Multi-Business Monitoring System with Sales Tracking, Inventory Management, and Financial Reporting for the Office of Business Affairs at Pampanga State Agricultural University is driven by the critical need to address prevailing operational challenges and align with the standards of public higher education institutions, particularly as the office manages diverse revenue-generating activities and multi-business operations including sales of agricultural produce, institutional supplies, assets, and services alongside the accountability of public funds and resources that require strict transparency and compliance. Presently, sales, inventory, and financial records are often managed through disjointed processes or separate systems creating information silos that hinder effective multi-business monitoring, leading to data discrepancies, redundant data entry, inconsistent performance views, and heavy administrative workload, which as noted by Juleb (2025) makes cross-verification of records slow and error-prone, thus an integrated platform is essential to establish a single source of truth that simplifies audits, ensures all units work with accurate and consistent information, and adheres to the accountability requirements set for state universities and colleges. This integration and multi-business monitoring capability significantly boosts operational efficiency by automating repetitive tasks such as transaction logging, stock level updates, and report generation, minimizing human error that often arises from manual processes; according to On Wave Group (2024),



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integrated systems streamline end-to-end workflows allowing organizations to accomplish more with fewer resources and less effort, freeing up personnel to focus on strategic and value-added activities instead of routine administrative work. Furthermore, the system provides real-time visibility into critical business metrics, a capability that is indispensable for sound decision-making and effective multi-business oversight, as Deloitte cited in ClickBalance (2025) emphasizes that nearly 45% of enterprises lose potential opportunities due to lack of timely access to sales and inventory data, enabling the management of Pampanga State Agricultural University to respond promptly to operational needs, adjust procurement and sales strategies, and avoid costly issues such as stockouts or overstocking that could waste institutional funds.

Effective inventory control is also greatly enhanced, as poor stock management can compromise up to 30% of an organization's working capital as stated by the National Inventory Association cited in ClickBalance (2025), with the system automatically tracking stock movements across multiple business units, triggering alerts for low inventory levels, and ensuring accurate valuation of both consumable supplies and capital assets, optimizing resource utilization and reducing unnecessary costs.

In terms of financial management, the integrated multi-business monitoring platform ensures accurate, reliable, and compliant reporting by automatically syncing operational data with accounting records, eliminating the need for tedious manual reconciliation and reducing errors in financial statements that are vital for compliance with Commission on Audit (COA) requirements and government accounting standards; ContinuSys (2024) adds that such systems are equipped with built-in compliance features including audit trails, electronic approvals, and version controls that make meeting regulatory and accounting standards much easier, supporting the transparency and accountability required for public institutions.

Finally, the web-based structure ensures scalability and long-term value for multi-business operations, as DevRova (2025) explains that managing a single integrated platform is more cost-effective and easier to expand compared to maintaining multiple disconnected software solutions, allowing the Office of



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	Business Affairs at Pampanga State Agricultural University to adapt to growing operational demands, support the university's expanding programs and revenue-generating initiatives, and sustain efficient and transparent management of institutional resources over time.
Objectives	General Objective To develop, implement, and evaluate a Web-Based Multi-Business Monitoring System with Inventory and Financial Reporting for the Office of Business Affairs (OBA) at Pampanga State Agricultural University (PSAU) that transforms and streamlines the office's diverse revenue-generating operations by integrating real-time sales tracking, inventory management, and financial reporting capabilities – ultimately enhancing operational efficiency, financial transparency, and service delivery to the university community. Specific Objectives 1. To develop a web-based sales transaction module for recording and monitoring daily business operations. 2. To create an inventory management module that tracks stock levels in real time and generates low stock rates. 3. To design a financial reporting module capable of generating accurate and timely reports. 4. To centralize sales, inventory, and financial data into one integrated database system. 5. To improve operational efficiency by reducing manual processing and redundant encoding. 6. To evaluate and developed system in terms of functionality, usability, reliability, efficiency, and security using ISO 25010.
Significance of the Study	Significance of the Study The development and implementation of this system hold substantial value for the institution, its personnel, and its stakeholders. The study is deemed significant because it addresses critical operational challenges and supports the mandate for transparency, efficiency,



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	and accountability in managing public funds and resources. The beneficiaries of this study are as follows: 1. Office of Business Affairs (OBA) This is the primary beneficiary of the study. The system will: Streamline Operations: It will eliminate disjointed processes and manual recording, replacing them with an integrated platform that automates transaction logging, inventory updates, and report generation. This significantly reduces administrative workload, redundant tasks, and human error. Ensure Data Integrity: By establishing a "single source of truth," the office will have consistent, accurate, and reliable data across all business units, simplifying cross-verification, audits, and decision-making. Enhance Compliance: The built-in features such as audit trails, automated logs, and standardized reporting ensure strict adherence to the requirements of the Commission on Audit (COA) and government accounting standards, thereby strengthening transparency and accountability in managing public resources. 2. Pampanga State Agricultural University (PSAU) Administration The system provides the university management with: Holistic Monitoring: A centralized view of all revenue-generating activities and multi-business operations allows the administration to monitor performance, income streams, and resource utilization in real time. Informed Decision-Making: Access to timely and accurate financial and operational data enables top management to formulate better policies, adjust strategies, and allocate resources efficiently to support the university's academic and operational goals. Digital Transformation: The project supports the university's compliance with Republic Act No. 11032 (E-Governance Act), helping the institution align with national standards for digitalization and modernization of public services. 3. Personnel and Staff The system directly benefits the users who handle daily operations by:
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	Reducing Workload: Automation of repetitive tasks frees personnel from tedious manual encoding and reconciliation, allowing them to focus on strategic and value-added activities. Improving Work Efficiency: Real-time data processing and retrieval speed up service delivery and transaction processing, resulting in a more productive and less stressful working environment. Ensuring Data Security: Role-based access control ensures that staff members only access information relevant to their duties, protecting sensitive data while ensuring smooth workflow. 4. Students and the University Community As the end-users and stakeholders of the university's services, the community benefits from: Better Resource Management: Effective inventory and financial management ensure that supplies, services, and facilities are adequately maintained and available when needed. Efficient Service Delivery: Streamlined operations result in faster transactions and improved services regarding institutional sales, rentals, and other revenue-generating activities. Transparency: Sound financial management ensures that funds generated are properly accounted for and utilized to improve facilities, academic programs, and student services. 5. Future Researchers and System Developers This study serves as a valuable reference and framework for future academic and technical studies by: Providing a Proven Framework: It documents the process of designing, developing, and implementing a web-based integrated system tailored for educational institutions, specifically those managing multiple revenue streams. Contributing to Knowledge: The findings regarding system performance, user acceptance, and operational efficiency can serve as a baseline for further research or development of similar systems in other state universities and colleges. Offering a Scalable Model: The modular and open-source nature of the system provides future developers with a solid foundation that can be enhanced, expanded, or adapted to suit other organizational needs.
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1. Integration of Sales, Inventory, and Financial Systems

The integration of operational and financial functions has been widely recognized as a critical strategy to enhance organizational performance. According to SAP (2025), linking sales, inventory, and accounting processes creates a seamless flow of information where every transaction is recorded once and reflected across all modules, eliminating the need for manual reconciliation and reducing data discrepancies. This "single source of truth" ensures that managers have consistent and accurate data for analysis, as highlighted by Juleb (2025), who notes that disconnected systems often lead to slow verification processes and errors that complicate audits and decision-making.

In the context of public sector management, integrated systems play an even more vital role. Research by ResearchGate (2025) emphasizes that combining financial management with inventory control enables real-time tracking of assets, budget allocations, and expenditures, minimizing errors and preventing fraud. Such systems foster transparency, streamline administrative workflows, and enhance compliance with regulatory requirements, which are essential for institutions managing public funds. Similarly, the Integrated Financial Management Information System (IFMIS) implemented in various government agencies demonstrates how unified platforms improve budget execution, procurement, and reporting, ensuring that resources are utilized efficiently and accountably (National Treasury, 2026).

For small and medium enterprises and institutional units, web-based solutions have proven effective in addressing operational challenges. Studies such as Risandi et al. (2025) and Semantic Scholar (2024) show that systems integrating sales monitoring, inventory management, and financial reporting reduce human error, provide real-time visibility of stock levels and financial performance, and enable faster generation of reports. These systems are particularly beneficial for organizations with diverse product lines and high transaction volumes, as they automate routine tasks and allow staff to focus on strategic activities.

2. Web-Based Systems in Higher Education Institutions

The adoption of web-based management information systems has become increasingly common in higher education institutions as a solution to streamline administrative processes. Abdullahi et al. (2024) found that traditional manual and paper-based operations often result in inefficiencies, delays, and limited accessibility, while online platforms centralize data and allow authorized users to access information anytime and anywhere. Fatoni et al. (2025) add that web-based systems digitize administrative tasks, improve information sharing among stakeholders, and enable faster and more accurate decision-making.



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In the Philippines, state universities and colleges are mandated to adopt electronic governance systems in line with the E-Governance Act (Republic Act No. 11032), which requires the integration of operations and compliance with national interoperability standards. Studies show that many institutions are now transitioning to digital platforms, with an assessment by Quodala (2022) revealing that 43% of universities have fully applied digitalization in their processes, while others are in various stages of implementation. Initiatives such as PHLChedConnect and Hi-Ed Bayanihan further support this shift by providing infrastructure and capacity building for digital transformation (Sinag Solutions, 2024).

Web-based ERP systems for education, as discussed by IJCRT (2025) and Cyborg ERP (2026), typically include modules for finance, inventory, human resources, and student management. These systems are designed with role-based access control to ensure data security, scalability to accommodate growing needs, and user-friendly interfaces that require minimal technical expertise. For institutions like Pampanga State Agricultural University, which manages both academic and revenue-generating activities, such integrated platforms are essential to maintain efficiency, transparency, and compliance with government standards.

3. Benefits of Automation and Real-Time Data

Automation is a key feature of modern integrated systems, offering significant improvements in speed and accuracy. According to On Wave Group (2024), automating tasks such as transaction recording, inventory updates, and report generation reduces processing time and minimizes errors that often occur in manual work. This is supported by studies on small business systems, where web-based solutions have been shown to increase operational efficiency by up to 60% and reduce administrative workload significantly (RSIS International, 2025).

Real-time data access is another critical advantage. Deloitte (cited in ClickBalance, 2025) notes that nearly 45% of organizations lose opportunities due to lack of timely information, while integrated systems allow managers to monitor sales trends, stock levels, and financial performance instantly. In the context of inventory management, this capability helps prevent overstocking or shortages, which can tie up working capital or result in lost revenue. The National Inventory Association (cited in ClickBalance, 2025) estimates that poor inventory management can affect up to 30% of an organization's capital, making real-time tracking essential for cost control.

In financial management, automation ensures that transactions are properly recorded and reflected in accounting records immediately. This simplifies the preparation of financial statements and supports compliance with standards set by the Commission on Audit (COA) and other regulatory bodies. Humedit Philippines (2025) highlights



that web-based systems with built-in audit trails and electronic approvals make it easier to maintain accurate, auditable records, which are required for public accountability and transparency.

4. Challenges and Considerations in System Implementation

Despite the benefits, implementing integrated systems also presents challenges. Research indicates that some organizations face difficulties related to infrastructure, technical skills, and user adaptation. Fatoni et al. (2025) note that while web-based systems offer many advantages, their success depends on reliable internet connectivity, adequate hardware, and training for staff to ensure proper use.

Another challenge is the complexity and cost of some existing solutions. Many advanced ERP systems are designed for large organizations and may be too expensive or complicated for smaller institutions or units. This has led to a growing demand for lightweight, affordable, and easy-to-use platforms that can be customized to specific needs (International Burch University, 2025).

Data security and privacy are also critical considerations. Studies by Adesemoye et al. (2021) and Mokogwu et al. (2024) emphasize the importance of features such as encryption, multi-factor authentication, and role-based access control to protect sensitive information. Web-based systems must also comply with data protection laws and regulations, ensuring that information is stored and processed safely.

For educational institutions, additional factors include the need to align systems with government requirements, integrate with existing processes, and ensure that the solution is sustainable in the long term. Scalability is important to allow for future expansion, while usability ensures that the system can be adopted effectively by all users regardless of their technical background.

5. Synthesis and Relevance to the Present Study

The reviewed literature clearly demonstrates the value of integrating sales, inventory, and financial functions into a single web-based system. Studies consistently show that such solutions improve efficiency, reduce errors, enhance transparency, and support better decision-making. In the context of higher education and public institutions, these systems are not only beneficial but also necessary to meet regulatory requirements and ensure accountability in the management of resources.

While many systems have been developed, there remains a need for solutions that are specifically tailored to the needs of institutions like Pampanga State Agricultural University, which manage diverse operations including agricultural produce, supplies, and services. The present study aims to address this gap by developing a web-



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based integrated system that combines sales monitoring, inventory control, and financial reporting in a platform that is affordable, user-friendly, and compliant with government standards. By leveraging modern web technologies and best practices from existing research, the system is expected to significantly improve the operations of the Office of Business Affairs and contribute to the overall efficiency and transparency of the university.

6. Multi-Business Monitoring and Centralized Management

Managing separate business operations within a single organization presents unique challenges in monitoring and control. According to Process Point (2025), without a centralized system, each unit tends to develop its own processes and data repositories, creating information silos that hinder effective oversight. A multi-business monitoring system resolves this by offering a centralized view while maintaining the ability to view data per specific unit or department. This structure ensures that while individual operations remain manageable, the administration has a holistic perspective of all activities and performance metrics.

Method CRM (2025) adds that centralized systems simplify administration by allowing updates and configurations to be done once and applied across all connected operations. This reduces the time and cost associated with maintaining multiple separate software solutions. Furthermore, these systems support role-based access control, meaning that specific personnel are granted permissions relevant to their duties—whether managing sales, inventory, or finance—ensuring both security and efficiency.

In higher education institutions where multiple revenue-generating centers operate, having a unified monitoring system is necessary to standardize processes. Hudutech Ventures (2025) explains that ERP systems designed for education sectors consolidate data from various departments, allowing the central administration to monitor income, expenses, and resource utilization consistently. This standardization is crucial in adhering to the financial policies and accounting standards required by regulatory bodies such as the Commission on Audit (COA).

7. System Development Technologies and Scalability

The choice of development tools and architecture determines the performance, security, and longevity of a system. DevRova (2025) notes that building a single integrated web-based platform is more cost-effective and easier to maintain than operating multiple separate systems. Web-based solutions developed using modern technologies like Node.js, Express.js, and Firebase offer robust security, flexibility, and scalability. According to Semantic Scholar (2024), cloud-based databases such as Google Firebase provide reliable data storage and accessibility, while security rules and authentication protocols ensure that sensitive financial and operational data remains protected. Technologies like JavaScript



and CSS enable the creation of responsive and user-friendly interfaces, making the system accessible and easy to learn for personnel with varying levels of technical expertise.

Rajan & Sulthana (2022) further note that modular system architecture allows for easy updates and feature additions as the organization grows. This scalability is vital for the Office of Business Affairs, as it allows the system to adapt to increasing transaction volumes, new business units, and evolving reporting requirements over time.

8. Integrated POS Systems in Educational Institutions

Educational institutions managing multiple revenue-generating units require systems that unify sales, inventory, and finance processes to ensure smooth operations and strict compliance. According to Asrani et al. (2024), traditional manual and disjointed systems remain prevalent in many public academic institutions, causing transaction delays, inventory discrepancies, and conflicting financial data that often lead to reporting errors and difficulty in audit preparation. This situation is evident in state universities where separate tracking for sales, supplies, and assets results in heavy administrative work and inconsistent records.

A study by Bautista & Dela Cruz (2023) on school-based enterprises revealed that implementing a unified POS system improved transaction speed by up to 50% and reduced data entry errors significantly. They emphasized that integrating sales recording with stock monitoring allows institutions to maintain accurate records of available resources, which is critical in managing both consumable supplies and fixed assets. Similarly, CPU Research and Development Center (2025) noted in their study on computerized sales and inventory systems that centralizing data enables administrators to monitor income streams from different units in real time, ensuring that all resources are properly accounted for and utilized.

In the Philippine setting, Mendoza et al. (2024) highlighted that for government educational institutions, integrated systems are not just operational tools but also instruments of accountability. These systems provide clear audit trails, trace every transaction, and generate standardized reports that align with the requirements of the Commission on Audit (COA). This makes the system vital in demonstrating transparency and proper stewardship of public funds and resources.

9. Real-Time Inventory Management and Control

Effective inventory management is one of the core functions that an integrated system addresses. Gill (2021) stated that real-time inventory tracking automatically updates stock levels as sales are processed, eliminating the need for manual stock counts and



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reducing the risk of both stockouts and overstocking. For institutions managing various items—from agricultural produce to office supplies—this capability ensures that resources are always available when needed while preventing excess stock that ties up capital. AlphaTechsUSA (2026) reported that organizations using integrated inventory systems reduce overstocking costs by approximately 10% and cut instances of stock shortages by 15%. This directly impacts financial performance by optimizing working capital and ensuring that funds are used efficiently. Furthermore, Cruz & Reyes (2019) explained that proper inventory management supported by technology helps organizations determine the movement of items, identify fast-moving and slow-moving goods, and make data-driven decisions regarding procurement and pricing strategies. In the context of public management, Eclarin & Dagamac (2018) emphasized that accurate inventory data is essential for resource planning and budgeting. When institutions have clear visibility of what they have on hand, they can minimize unnecessary purchases and allocate funds to more critical areas, improving overall financial sustainability.

10. Financial Reporting Automation and Compliance

Financial reporting is a rigorous process that demands accuracy, timeliness, and adherence to government standards. Rahmadani (2025) observed that manual preparation of financial statements is not only time-consuming but highly prone to human error, which can result in financial misstatements and non-compliance with regulatory requirements. An integrated system resolves this by automatically compiling data from sales and inventory operations into accounting records, ensuring that figures are consistent and verifiable.

Software 4 Schools (2025) pointed out that modern systems can generate different types of reports—such as sales summaries, cost of goods sold, income statements, and inventory valuation reports—with just a few clicks. These reports can be customized to meet the specific reporting formats required by government agencies, simplifying the auditing process and ensuring that all financial activities are documented correctly.

Eezipay (2026) added that automation also strengthens financial security and accountability. By recording every transaction with timestamps and user identification, the system creates an immutable log that prevents unauthorized alterations and makes it easy to trace any discrepancies or irregularities. This level of control is essential for state universities, which are subject to strict financial regulations and public scrutiny.

11. Web-Based Architecture and Accessibility

The use of web-based technology offers distinct advantages suitable for institutional management. SalesPlay (2025) explained that unlike traditional desktop applications, web-based systems allow authorized users to access data and perform tasks from different locations and devices, provided there is an internet connection. This



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	accessibility is particularly useful for the Office of Business Affairs, where personnel may need to monitor operations or process transactions across different units or campuses. ZBS POS (2025) noted that cloud-based web systems also simplify maintenance and data security. Data is stored securely on remote servers with regular backups, eliminating the risk of data loss due to hardware failure or physical damage. Updates and improvements can also be deployed centrally, ensuring that all users are working with the latest and most secure version of the system. Sia & Torres (2023) further emphasized that web-based systems are highly scalable. As the university expands its programs, introduces new revenue streams, or increases transaction volumes, the system can easily be expanded or upgraded without requiring a complete overhaul of the infrastructure. This long-term adaptability makes it a cost-effective solution that supports the growth and development goals of the institution. 12. Operational Efficiency and Administrative Workload Reduction One of the primary motivations for system development is to streamline operations and reduce manual workload. On Wave Group (2024) previously noted that automation eliminates redundant tasks, allowing staff to shift their focus from repetitive administrative work to more strategic and value-adding activities. ZCS (2025) supported this, stating that integrated systems can reduce the time spent on data entry, reconciliation, and reporting by as much as 40–60%. In the study conducted by Bautista & Dela Cruz (2023), staff members reported significant improvements in their work performance after system implementation, citing reduced stress, faster turnaround time, and improved accuracy in their outputs. This improvement in workflow efficiency also translates to better service delivery, as transactions are processed faster and inquiries regarding sales, stock, or finances can be answered immediately with accurate data. ClickBalance (2025) reiterated that efficiency gained through automation directly contributes to better financial health, as resources are utilized more effectively, costs are controlled, and management has the necessary tools to make timely and informed decisions that benefit the entire organization.
Methodology	7.1 Research Design This study employs the Descriptive-Developmental Research Design, a two-pronged approach that combines descriptive inquiry with system development activities. This design is specifically chosen because it aligns perfectly with the objectives of the study, which involve both understanding the existing situation and creating a solution tailored to the identified needs.



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The descriptive component serves as the foundation of the study. It involves gathering detailed information about the current operational processes of the Office of Business Affairs (OBA) at Pampanga State Agricultural University. Through observations, interviews, and document reviews, this phase aims to describe the status quo — including how sales transactions are recorded, how inventory items are monitored and replenished, and how financial data is consolidated and reported. It also seeks to identify existing challenges, limitations, and inefficiencies, such as manual recording errors, delayed reporting, lack of real-time data access, or difficulties in tracking inventory movement. Additionally, this phase determines the exact functional and non-functional requirements that the proposed system must satisfy to effectively address these issues. By clearly defining the current state and the desired outcomes, the descriptive phase ensures that the system to be developed is relevant, necessary, and responsive to actual operational needs.

The developmental component, on the other hand, focuses on the actual creation and implementation of the web-based integrated system. Using the findings from the descriptive phase as a blueprint, this part of the study follows a systematic process to design, build, test, and deploy a system that integrates sales monitoring, inventory management, and financial reporting functionalities. The goal is to transform the identified requirements into a fully functional, usable, and reliable system that streamlines operations, improves data accuracy, and enhances decision-making processes within the OBA. This design is appropriate for the study because it not only answers the question “what is happening?” but also addresses “how can we improve it?” by producing a tangible solution based on empirical data.

7.2 System Development Life Cycle

To ensure that the system is developed in a systematic, organized, and high-quality manner, the study follows a structured framework known as the System Development Life Cycle (SDLC). This framework provides a clear roadmap, dividing the complex process of system development into manageable phases, each with defined objectives, activities, and deliverables. Adhering to an SDLC ensures that all aspects of the system — from initial planning to final deployment — are thoroughly considered, documented, and validated, reducing risks and increasing the likelihood of project success.

7.2.1 System Development Life Cycle Model



For this study, the V-Model (Verification and Validation Model) is adopted as the specific SDLC framework. Unlike linear models where testing is done only after development is complete, the V-Model maps each development phase to a corresponding testing phase, forming a “V” shape. This means that planning and design activities are immediately followed by relevant testing preparations, and every stage of development is verified before moving to the next.

This model was selected for several important reasons:

- **Emphasis on Quality Assurance:** Since the system deals with financial and inventory data information that requires high levels of accuracy, reliability, and security it is critical to detect and correct errors as early as possible. The V-Model ensures that verification and validation are integral parts of the process, not just afterthoughts.
- **Clear Alignment:** Each development phase has a corresponding testing phase, which ensures that requirements are translated correctly into design, and design into code. This alignment guarantees that the final product truly meets the specified needs of the OBA.
- **Structured Documentation:** The model requires detailed documentation at every stage, which is essential for academic research purposes, as it allows for transparency, reproducibility, and thorough evaluation by stakeholders and panelists.
- **Defined Milestones:** The V-Model establishes clear deliverables and checkpoints, making it easier to monitor progress, manage resources, and ensure that the project stays within scope and schedule.

Given the nature of the system which supports core business operations and financial reporting the V-Model is the most suitable choice because it prioritizes correctness, stability, and compliance with user requirements throughout the entire development process.

7.2.2 Development Phases

- The V-Model is divided into two distinct but interconnected parts: the development phases (left side of the V) and the testing/validation phases (right side of the V), culminating in deployment and evaluation. Each phase is explained in detail below:

Development Phases (Left Side of the V)



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	- These phases focus on defining, designing, and building the system, moving from general concepts to specific technical implementation. • Planning & Requirements Analysis - This is the initial and most critical phase, as it lays the groundwork for the entire project. The researchers engage in close collaboration with OBA personnel — including managers, accountants, inventory officers, and encoding staff to gather comprehensive information about their operations. Through interviews and observations, the team identifies what the system should do (functional requirements) such as recording sales, updating inventory, and generating financial reports, as well as how it should perform (non-functional requirements) such as security, speed, ease of use, and accessibility. Additionally, this phase documents existing problems, constraints, and expectations. The output of this phase is a detailed Requirements Specification Document, which serves as the primary reference for all subsequent development activities. This ensures that the system is built based on actual user needs and not just assumptions. • System Design - Once the requirements are clearly defined, the system design phase translates these requirements into a logical structure and visual blueprint. This involves creating the overall system architecture, which shows how different components and modules will interact with each other. It also includes designing the user interface (UI) layouts determining the arrangement of menus, forms, buttons, and displays to ensure that the system is intuitive and user-friendly. Process flow diagrams are also developed to illustrate the sequence of operations, data movement, and decision points within the system. The goal of this phase is to provide a clear, high-level representation of the system's structure and behavior before actual coding begins. • Detailed Design - While system design focuses on the big picture, detailed design breaks down the system into specific technical specifications. This phase involves designing the database schema — defining tables, fields, data types, and relationships to ensure that data is stored efficiently, accurately, and securely. It also outlines the internal logic and algorithms for each module: for example, how sales data will
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	be computed, how inventory levels will be adjusted, or how financial summaries will be generated. Special attention is given to the three core modules of the system: Sales Monitoring, Inventory Management, and Financial Reporting, ensuring that each is designed to perform its intended functions seamlessly and can integrate smoothly with the others. The deliverable of this phase is a Technical Design Document that guides the coding process. • Implementation & Coding - In this phase, all the designs and specifications are transformed into actual software using the selected programming languages and tools. The developers write the source code for each module, following best practices in coding standards and structure. Features are built one by one, starting from basic functions to more complex operations. As each part is completed, it is integrated with other components to form a unified system. This phase also includes setting up the database, configuring the server environment, and ensuring that all parts of the system communicate correctly. The result of this phase is a fully working but untested version of the system. Testing & Validation Phases (Right Side of the V) These phases run parallel and correspond to the development phases, ensuring that every aspect of the system is checked and verified against the requirements and designs. • Unit Testing - Corresponding to the Detailed Design phase, unit testing involves testing individual components, functions, or modules in isolation. Each piece of code is checked to ensure that it performs exactly as specified in the technical design. For example, a function that calculates total sales or updates inventory counts is tested with different inputs to confirm it produces the correct output. This phase helps detect errors at the smallest level, making them easier and less costly to fix. It ensures that each building block of the system is reliable before being combined with others. • Integration Testing - Corresponding to the System Design phase, integration testing focuses on how different modules and components work together. Once individual units are tested and approved, they are combined, and the interactions between them are verified. This phase checks if data flows correctly from one module to another, if the database is accessed properly, and if there are no conflicts or communication errors between
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components. For instance, it verifies whether data entered in the Sales module is correctly reflected in the Inventory module and eventually included in Financial Reports. This ensures that the system works as a cohesive whole rather than just a collection of separate parts.

- **System Testing**

- Corresponding to the Planning & Requirements Analysis phase, system testing evaluates the complete and integrated system against all the requirements defined at the start. It covers all aspects of the system: functionality (does it do what it is supposed to do?), performance (is it fast enough even with large amounts of data?), security (is data protected from unauthorized access?), usability (is it easy to learn and use?), and reliability (does it operate consistently over time?). This phase simulates real-world usage scenarios to ensure that the system meets all the needs and expectations outlined in the Requirements Specification Document.

- **User Acceptance Testing (UAT)**

- This is the final testing phase and is conducted by the actual end-users — the OBA staff and administrators — rather than just the developers or researchers. Users are given the system to perform their regular tasks and operations, allowing them to test it in a real operational environment. They assess whether the system is suitable for daily use, if it solves their previous problems, and if it meets their operational needs. Feedback from UAT is crucial: if users identify issues or suggest improvements, these are addressed before the system is officially deployed. Only when users confirm that the system is acceptable and ready for use does the project move to the next stage.

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Deployment & Evaluation

After successfully passing all testing phases, the system is officially deployed or installed for regular use. This includes setting up the system in the OBA's working environment, migrating existing data to the new system, and providing comprehensive training to users so they are confident and capable of operating it effectively. Following deployment, a post-implementation evaluation is conducted to assess the system's actual performance, usability, and impact on daily operations. This phase ensures that the system continues to function correctly and delivers the intended benefits over time.

7.3 Methods of Data Collection



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To gather accurate, relevant, and comprehensive data for both the descriptive phase (requirements gathering) and the developmental phase (system evaluation), the study utilizes the following methods:

- **Interview**

Interviews are conducted with key personnel from the OBA, including office heads, supervisors, and frontline staff. This method allows the researchers to have direct, in-depth conversations with users, enabling them to ask open-ended questions and explore topics in detail. Through interviews, the team gathers information about current workflows, specific difficulties encountered in daily tasks, and their expectations and preferences regarding the proposed system. Interviews are valuable because they provide qualitative insights that may not be captured through observation or document analysis alone. They also help build a collaborative relationship between the researchers and the users, ensuring that the system is truly user-centered.

- **Observation**

Direct observation involves the researchers watching and recording the actual daily operations of the OBA related to sales, inventory, and financial reporting. By seeing how tasks are performed in real life such as how sales transactions are recorded, how stock levels are checked, or how reports are prepared the researchers can identify processes that are inefficient, redundant, or prone to error. Observation is particularly useful because it allows the team to verify whether what is said in interviews matches what is actually done in practice. It also helps uncover unspoken needs or hidden challenges that users may not explicitly mention.

- **Document Analysis**

This method involves reviewing and examining existing documents and records used by the OBA, such as sales logs, inventory ledgers, financial statements, transaction receipts, and existing report formats. Analyzing these documents provides concrete evidence of the data being collected, the formats currently used, and the types of information required for decision-making and reporting. It also helps identify inconsistencies, missing information, or areas where data could be better organized or utilized. Document analysis ensures that the proposed system is designed to handle the same types of data and reports that the office is already using, making the transition to the new system smoother and more effective.

By combining these three methods, the study ensures that data is gathered from multiple perspectives, leading to a more complete and accurate understanding of the situation and resulting in a system that is well-aligned with the actual needs of the OBA.



7.4 Respondents of the Study

The respondents of the study are the staff and personnel of the Office of Business Affairs at Pampanga State Agricultural University. This group is selected because they are the primary stakeholders and end-users of the proposed system. They are directly involved in the core operations covered by the system sales monitoring, inventory management, and financial reporting and therefore possess the necessary knowledge, experience, and understanding of the processes, challenges, and requirements related to these tasks.

The respondents include personnel from different roles and levels within the office, such as:

- Administrative and encoding staff who perform day-to-day data entry and recording
- Inventory officers responsible for monitoring stock levels and movement
- Accounting and finance staff who prepare financial reports and analyze data
- Office heads or supervisors who oversee operations and make decisions based on the reports generated

By involving users from various roles, the study ensures that all perspectives and needs are considered, leading to a system that caters to everyone who will use it. Their participation is essential not only in defining what the system should do but also in evaluating whether the final product is effective, usable, and beneficial to their work.

7.5 Data Analysis Procedures

Different analysis procedures are used depending on the purpose of the data collected:

- **For Requirements Gathering**

The data collected from interviews, observations, and document analysis is analyzed using Thematic Analysis. This is a qualitative method of analysis that involves organizing and interpreting data to identify common themes, patterns, and recurring issues. The process involves transcribing interviews, summarizing observation notes, and reviewing document contents. Similar ideas, problems, or suggestions are grouped together into themes — for example, “difficulty in tracking inventory in real-time” or “need for automated financial summaries.” These themes are then interpreted to understand their meaning and implications for the system design. The result of this analysis is a clear set of system requirements derived directly from the experiences and needs of the users.

- **For System Evaluation**



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Once the system is developed and tested, quantitative methods are used to assess its quality and performance. Descriptive statistics are applied to the data gathered from evaluation surveys or feedback forms administered to the respondents. The following statistical tools are used:

- Frequency Count: To determine how many respondents gave a particular rating or response.
- Percentage: To show the proportion of respondents who agree or disagree with certain statements or criteria.
- Weighted Mean: To calculate the average rating given by respondents, which reflects the overall level of satisfaction or performance of the system.

The system is evaluated based on standard criteria for information systems, including:

- Functionality: Whether the system performs all required tasks accurately and completely.
- Efficiency: How fast the system operates and how well it uses resources.
- Usability: How easy and intuitive the system is to learn and use.
- Reliability: Whether the system operates consistently and correctly over time and under different conditions.

The results of the analysis are presented in tables and narrative form to clearly describe the system's strengths and areas for improvement.

7.6 Tools and Technologies

The following tools and technologies are selected based on their suitability for the project requirements, ease of use, reliability, and ability to support the desired features of the system. Each tool is explained below, including why it was chosen and how it is used in the development process.

Personal Computer Specifications

- **Operating System: Windows 11**
- **Memory (RAM): 16GB**
- **Storage Capacity: 1TB**
- **Graphics Processing Unit: Integrated Graphics Processing Unit (IGPU)**

These specifications were chosen to provide a stable, efficient, and capable environment suitable for the entire system development lifecycle. Windows 11 is selected as the operating system because it offers a modern, secure, and user-friendly platform that is fully



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compatible with all the development tools, frameworks, and applications used in this study. It also includes built-in security features, regular updates, and improved performance optimizations that help ensure smooth development operations.

- **Code Editor: Visual Studio Code (VS Code)**

VS Code is the primary environment used for writing, editing, and managing all the source code of the system. It was chosen because it is lightweight yet powerful, supports multiple programming languages, and has built-in features like syntax highlighting, error detection, and debugging tools that make development faster and more efficient. It also supports extensions that allow developers to customize the environment according to their needs, making it ideal for building web-based systems.

- **Backend Technology: Node.js**

Node.js is the runtime environment used to build the server-side or backend part of the system — the part that handles data processing, business logic, and communication between the user interface and the database. It is chosen because it allows developers to use JavaScript (a language familiar to many developers) for both the frontend and backend, making the development process more consistent and easier to manage. It is also known for being fast, scalable, and capable of handling multiple requests at the same time, which is important for a system that may be accessed by several users simultaneously.

- **Security: Helmet**

Helmet is a security middleware for Node.js applications. It is used to enhance the security of the system by automatically setting secure HTTP headers, which help protect the system from common web vulnerabilities such as cross-site scripting (XSS), clickjacking, and other attacks. Since the system handles sensitive financial and operational data, ensuring strong security is a top priority, and Helmet provides a simple but effective way to implement essential security measures.

- **System Structure: Express.js & HyperText Markup Language (HTML)**

Express.js is a lightweight framework built on top of Node.js that simplifies the process of organizing the backend structure and managing routes or URLs. It helps organize the code in a logical way, making it easier to maintain and update. HTML, on the other hand, is the standard language used to create the structure and content of the web pages or user interface. It defines the elements that users see and interact with, such as forms, tables, buttons, and menus. Together, these tools provide a solid foundation for building a well-structured web application.



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	• Styling: Vanilla Cascading Style Sheets (CSS) CSS is used to control the visual appearance and layout of the system interface. It defines how elements are displayed including colors, fonts, spacing, alignment, and positioning. Using “vanilla” or plain CSS (without additional frameworks) allows the developers to have full control over the design, ensuring that the interface is customized specifically to meet the needs and preferences of the OBA users. It also ensures that the system is lightweight and loads quickly. • Interactivity: JavaScript JavaScript is the programming language used to add dynamic behavior and interactivity to the system. It enables features such as real-time calculations, form validation, dynamic content updates, and interactive elements that respond to user actions all without needing to reload the entire page. This improves the user experience by making the system more responsive and easier to use. JavaScript
Expected Output (Proposed Features and Fucntionality of the product)	The expected output of this capstone project is a fully deployed, web-based integrated management system for the Office of Business Affairs (OBA) and all the business connected to the Office of Business Affairs (OBA) at Pampanga State Agricultural University. The system is designed to centralize and automate sales monitoring, inventory management, and financial reporting operations through a secure, role-based, and real-time platform accessible via standard web browsers. The following are the proposed modules, features, and functionalities of the OBA System: Module 1: Authentication and User Access Management This module governs who can access the system and what they are permitted to do. 1.1 Role-Based Login System • Secure login and logout functionality for all authorized personnel. • User roles include: System Administrator, Accounting Officer, Budgeting Officer, and Sales Staff - each with defined access permissions. • Firebase Authentication is used to manage credentials, with session persistence and token-based verification.



	1.2 Access Request Workflow - Accounting and Budgeting offices must request access through a formal workflow approved by the OBA System Administrator. - Administrators can grant, modify, or revoke user roles at any time through the Admin Panel. 1.3 Professional OBA Login Page - A branded, responsive login interface displaying the OBA System logo, system name, and a user-friendly credential form. - Input validation and error handling for incorrect credentials or unauthorized access attempts. Module 2: Sales Monitoring and Reporting This module enables authorized personnel to record, view, filter, and generate reports on all sales transactions of the Office of Business Affairs. 2.1 Real-Time Sales Overview Dashboard - A live dashboard displaying total sales for the current day, week, and month. - Key metrics include: total revenue, number of transactions, average sale value, and top-selling products. - Data is retrieved and updated in real time from Firebase Firestore. 2.2 Sales Transaction Recording - Authorized staff can record new sales transactions capturing: Transaction ID, date and time, items sold, quantities, unit prices, total amount, and payment method (cash, credit, GCash, etc.). - Automatic computation of transaction totals, payment change, and balance upon entry. - Transactions are stored immediately to Firestore with a system-generated unique ID and timestamp. 2.3 Daily and Monthly Sales Viewing and Printing - Users can view a complete list of all sales transactions filtered by day, week, or month. - Sales records can be printed in a clean, formatted layout for documentation and compliance purposes.
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	• Supports export to printable report format with system header, date, and prepared-by fields. 2.4 Customizable Sales Filters and Billing • Filter sales data by: payment type (cash, online payment), product category, date range, and staff member. • Generate filtered sales reports (e.g., 'Monthly Credit Sales Report'). • Billing generation feature for individual transactions for client documentation. Module 3: Smart Inventory Management This module tracks and manages the stock levels of all products handled by the Office of Business Affairs in real time. 3.1 Product Record System • Maintains a comprehensive product master list with fields: product name, description, category, unit of measure, unit cost, and supplier information. • Products can be added, updated, or archived by authorized users. 3.2 Real-Time Stock Tracking • Every sales transaction automatically deducts the corresponding product quantity from the Firestore inventory collection. • Every restocking entry updates stock levels in real time. • Inventory status (In Stock, Low Stock, Out of Stock) is displayed prominently per product. 3.3 Automated Low-Stock Alerts • Firebase Cloud Functions monitor Firestore for products falling below a predefined reorder threshold. • Immediate on-screen notifications are triggered when stock levels reach the alert threshold. • Alert history is logged in the system for audit purposes. 3.4 Product Restock Notifications • When new stock is received and recorded, the system automatically updates inventory totals and logs a restock event.
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	- Restock records include: date received, quantity added, supplier, and purchase cost. 3.5 Comprehensive Inventory Movement History - A complete audit trail of all inventory transactions: sales deductions, restocks, manual adjustments, and returns. - Filterable by product, date range, or transaction type for easy review and reporting. Module 4: Advanced Financial Reporting This module generates accurate, automated financial reports and analytical tools to support the financial decision-making and compliance requirements of the Office of Business Affairs. 4.1 Dynamic Report Generator - Automatically compiles and formats financial data into professional reports. - Available report types: Daily Sales Reports, Weekly Revenue Summaries, Monthly Financial Statements, Transaction History Reports, Expense Reports, and Profit & Loss Statements. - All reports include trend indicators (up/down) comparing the selected period to the previous period. - Reports are fully printable with professional layout, OBA branding, and auto-generated header fields. 4.2 Key Performance Indicator (KPI) Dashboard - A visual, customizable dashboard providing a high-level overview of: total income, total expenses, net profit/loss, inventory value, and sales performance by product category. - Interactive charts (bar, line, pie) powered by JavaScript charting libraries for intuitive data visualization. - Data is sourced directly from Firestore and updated in real time. 4.3 Automatic Financial Computation - The system automatically computes: total sales per transaction, change and payment balances, daily/weekly/monthly revenue aggregates, inventory cost and stock valuation, and profit and loss calculations. - Eliminates all manual computation, reducing errors and ensuring data integrity.
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4.4 Return on Investment (ROI) Calculations

- Built-in tools to calculate ROI for specific products or product categories.
- Users can apply date-range filters to generate ROI reports for specific campaigns or time periods.

Module 5: Centralized Data Management and System Administration

This module ensures that all data across sales, inventory, and finance is stored, maintained, and accessible from a single integrated platform.

5.1 Unified Firestore Database

- All modules — sales, inventory, financial records, user profiles, and system logs — are stored in a structured, centralized Firestore database.
- Real-time data synchronization ensures all users see the most current information at all times.

5.2 System Audit Trail and Logs

- Every user action (login, data entry, edit, deletion, report generation) is logged with a timestamp and user ID.
- Audit logs are accessible by the System Administrator for compliance with Commission on Audit (COA) requirements.

5.3 Automated Database Backup

- Scheduled automated backups of all Firestore data to prevent data loss.
- Backup history is retained and restorable by the system administrator.

5.4 Website Public Pages

- Home Page - Hero section with system tagline, About OBA, Mission, Vision, and Why OBA Matters highlights.
- Features Page - Detailed overview of all system modules.
- Contact / Request a Demo Page - Form for interested parties to request access or submit inquiries.
- Careers Page - For institutional staff recruitment communication.



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	5.5 Employee Pages - Assigned Business Dashboard: Provides a focused view of the specific business units (B1–B17) the employee is authorized to manage. - Sales Processing Terminal: A streamlined interface for recording daily sales transactions, featuring auto-computation of subtotal, tax, and change. - Inventory Management Tools: Allows employees to update stock levels, log received items from suppliers, and monitor reorder points for their specific business unit. - Transaction History: Real-time access to a searchable log of all sales recorded by the employee, with the ability to reprint digital receipts. - Operational Reports: View-only access to daily sales summaries and stock status reports to ensure smooth day-to-day operations. 5.6 View and Print only - Read-Only KPI Monitoring: Provides access to high-level system dashboards, including revenue charts and total transaction counts, without editing permissions.
Sustainability plan	The proposed system is designed to remain operational and beneficial long after the study concludes. This Sustainability Plan outlines the strategies, responsibilities, and measures that will ensure the system continues to serve the Office of Business Affairs effectively in managing sales, inventory, and financial reporting. 1. Technical Sustainability 1.1 Open-Source Software Development The system is built entirely on free and open-source technologies, including HTML, CSS, JavaScript, Node.js and Firebase/Firestore, meaning there are no licensing fees, no subscription costs, and no financial barriers to keeping it running over time. These tools are backed by large, active developer communities, so updates, security patches, and technical support will remain available long after the study ends without any added cost to the university. 1.2 Focus on Essential Features



1.2.1 The system is built around the features that directly solve the problems identified at the Office of Business Affairs, **namely sales** transaction recording, real-time inventory monitoring, stock alert notifications, financial report generation, and secure user access control. Keeping the scope focused means the system stays lightweight, stable, and easy to maintain without requiring a dedicated IT team to keep it running.

1.3 Web-Based and Cross-Platform Design

Recognizing that staff use various devices from desktop computers to laptops, the system is designed to be responsive and accessible via any standard web browser. It does not require high-end specifications to operate, ensuring compatibility with the existing hardware infrastructure of the university.

1.4 Scalability and Adaptability

The system is built modularly, meaning new features or adjustments can be added over time without disrupting what already works. Changes in university policies, reporting requirements, or expansion of business units can be accommodated without rebuilding the system from scratch. Feedback from the staff during daily use will guide any future improvements, keeping the system relevant as the needs of the Office of Business Affairs evolve.

1. Operational Sustainability

2.1 Training and Capacity Building

Hands-on training will be conducted for the staff of the Office of Business Affairs covering every module in the system — product encoding, sales processing, inventory adjustments, report generation, and user management. A printed user manual and video tutorials will be provided as permanent references so staff can operate the system independently even without the developers present. Refresher sessions will also be available to accommodate new employees or updates to system features.

2.2 Formal Handover

Upon completion of the study, all system components, source code, database files, documentation, and the user manual will be formally transferred to Pampanga State Agricultural University. The development team will provide technical support for six months following turnover for troubleshooting and minor updates. After this period, full operational responsibility rests with the Office of Business Affairs and the University ICT Office.



2.3 Responsible Parties

The development team is responsible for system design, development, testing, initial training, and technical support during the post-turnover period. The Head or designated staff of the Office of Business Affairs will serve as the primary system administrator, overseeing day-to-day operations, user account management, and data integrity. The University Information and Communications Technology Office (UICTO) will provide institutional oversight and technical assistance to ensure the system remains aligned with university standards. Should technical issues arise beyond the support period, the office may coordinate with the Department of Computer Studies for further assistance.

2.4 Monitoring and Evaluation

Periodic check-ins between the development team and the office staff will be conducted during the six-month support period to confirm that all modules are actively used and records are consistently updated. System usage logs will be reviewed to identify any gaps in adoption or workflow issues. After the support period, the office is encouraged to conduct annual internal reviews to assess data accuracy, user satisfaction, and compliance with accounting and auditing standards.

3. Economic Sustainability

3.1 Operational Cost Transparency

It would be easy to say that the system costs nothing to run because all the software behind it is open-source and free. That is technically true; there are no licensing fees, no paid frameworks, and no subscription charges tied to the code itself. But saying that and stopping there would leave the office unprepared for what comes after deployment, and that is not something the researchers are comfortable doing.

The honest answer is that running a live system has costs. They are not large, but they are real, and the people who will be depending on this system deserve to know about them from the beginning.

Keeping the system accessible requires server resources and domain maintenance. While the university may utilize its existing infrastructure, should external hosting be required, these are costs that need to be planned for annually to ensure uninterrupted service.

Cost Item	Type	Estimated Cost



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Server Hosting / Maintenance	Annual	Varies (Can Use existing University Server)
Domain Name Registration	Annual	500-1,500 Pesos/ year
Open Source Software Licenses	None	0.00 Pesos

Table 1: Deployment Cost Analysis

When these costs are weighed against what the office currently spends in man-hours, paper supplies, printing, and manual reconciliation, the math still works in the system's favor by a wide margin. The recommendation is simple — include these minimal operational costs in the annual budget proposal so that a system built to help the university can actually keep doing that long after the researchers have moved on.

3.2 Reduction of Operational Costs

By replacing paper-based ledgers, manual logbooks, and spreadsheets, the system reduces the time and resources spent on administrative tasks. Staff spend less time encoding and consolidating data, and more time on analysis. The risk of data loss and errors, which carries its own cost in correction time, is significantly reduced. The system also helps prevent financial losses due to inventory discrepancies or unrecorded sales.

3.3 Cost-Benefit Impact

Benefit	Impact
Reduced Manual Work	Saves hours spent on manual tallying and report preparation
Eliminated Paper Usage	Saves cost of forms, ledgers, printing, and physical storage
Real-time Stock Alerts	Prevents stockouts or overstocking that tie up capital



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Faster Financial Reports	Ensures timely submission and compliance with COA/CHED requirements
Data Accuracy	Reduces errors and risks associated with manual recording

Table 2: Detailed Cost Benefits

4. Environmental Sustainability

Shifting from paper-based records to a centralized digital system directly reduces the volume of physical forms, ledgers, and printed reports consumed by the office on a daily basis. Requisition forms, sales invoices, inventory cards, and financial statements that are currently printed and stored in bulk will be digitized. Since the system operates entirely on software with no special hardware requirements, it produces no electronic waste and has a minimal environmental footprint. This supports a "green" and sustainable approach to university administration.

5. Institutional Sustainability

The system will be formally turned over to the Office of Business Affairs as an institutional management tool, not simply a capstone deliverable. The staff will own and operate the platform as part of their regular workflow, ensuring continuity of use beyond the project period. Role-based access control ensures that only authorized personnel can manage sensitive data, maintaining institutional accountability long after the developers' involvement has ended. Future capstone or research teams may also build upon the system's existing database and modules to conduct further studies or expand its capabilities as the university grows.

6. Risk Assessment and Mitigation

Risk	Likelihood	Impact	Mitigation
Staff Turnover (Personnel leaving or retiring)	Medium	High	User manual and refresher training ensure new



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				staff can be onboarded independently.
	System Bugs or Errors	Medium	Medium	Six-month post-turnover technical support period provided by the development team.
	Data Loss	Low	High	Automated database backups are scheduled regularly to prevent permanent loss of records.
	Low Digital Literacy	Medium	High	Hands-on training provided; interface is kept simple and user-friendly.
	Hardware/Infrastructure Issues	Medium	Medium	System is web-based and lightweight, compatible with standard office computers.



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	Unauthorized Access	Low	High	ole-based access control, secure login authentication, and encrypted database storage.
	Budget Constraints	Low	Low	Core software is free; minimal costs are easy to integrate into the regular office budget.
Web-based Integrated Sales Monitoring, Inventory, and Financial Reporting System 1. System and Software Updates • Incremental System Improvements: The system architecture is designed to allow continuous refinement as more transaction data is accumulated, enhancing reporting accuracy and process efficiency over time. • Modular Software Upgrades: The codebase and database structure are built modularly, enabling feature enhancements, compliance updates, or functional expansions without requiring a complete system overhaul. • Remote Administration: As a web-based platform, administrators can monitor system performance, deploy patches, and update configurations remotely, ensuring high availability and minimizing downtime for the Office of Business Affairs. 2. User Training and Knowledge Transfer • Training Program: Staff will receive comprehensive training on system navigation, data encoding, report generation, and reconciliation procedures to ensure efficient daily operations. • Documentation: Detailed user manuals, step-by-step guides, and video tutorials will be provided to support independent troubleshooting and smooth onboarding of new personnel.				



	• Support Channel: A dedicated feedback mechanism will be established to address user concerns, gather suggestions, and share best practices among staff members. 3. Operational and Institutional Sustainability • Process Optimization: By integrating sales, inventory, and financial functions into one platform, the system reduces manual errors, eliminates redundant data entry, and ensures real-time data accuracy. • Low Maintenance Infrastructure: Utilizing standard web technologies ensures compatibility with existing institutional hardware, minimizing additional infrastructure costs and supporting long-term deployment. 4. Scalability and Future Expansion • Modular Design: The system structure allows for the easy integration of additional modules (e.g., asset management, procurement tracking, or online payment gateways) as the university's needs grow. • Adaptability: The database and logic are designed to accommodate increasing transaction volumes, new business units, or changes in accounting standards with minimal modification. • Institutional Adoption: The system is designed for ease of use and standardization, making it suitable for widespread adoption and sustainability as a permanent tool for the university. 5. Continuous Evaluation and Improvement • Performance Monitoring: System logs, response times, and data integrity will be reviewed periodically to ensure the system continues to meet the operational requirements of the Office of Business Affairs. • Iterative Enhancement: User feedback and observed operational bottlenecks will be used to guide regular updates, ensuring the system remains relevant and efficient. • Foundation for Digitalization: This system serves as a core component of the university's digital transformation, paving the way for future integration with other administrative and academic information systems. Conclusion
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	This plan ensures that the Web-based Integrated Sales Monitoring, Inventory, and Financial Reporting System remains reliable, secure, and efficient in the long run. By combining robust software design, comprehensive user training, and continuous improvement, the system effectively supports the Office of Business Affairs at Pampanga State Agricultural University in achieving transparent, accurate, and streamlined management of institutional resources.
Project Workplan	Pls. see attached form
References	Abdullahi, H. O., Mohamud, I. H., & Gele, A. O. M. (2024). Best practices for the implementation of online management systems in universities. Journal of Syntax Admiration. Aramis Solutions. (2026). Integrated ERP System for Finance, Inventory & Sales. Retrieved from https://www.aramissolutions.com/integrated-erp-system-for-finance-inventory-sales/ Asrani, K., et al. (2024). Challenges of Manual and Disjointed Business Systems in Public Educational Institutions. Journal of Information Systems and Management Research. Bautista, R., & Dela Cruz, M. (2023). Effectiveness of Computerized Sales and Inventory Systems in Managing School-Based Income-Generating Projects. Philippine Journal of Development and Governance. ClickBalance. (2025). La integración entre gestión comercial, inventario y finanzas: clave para la rentabilidad de tu PyME. Retrieved from https://clickbalance.com/blog/integracion-gestion-e-inventario ContinuSys. (2024). 7 Benefits of an Integrated Business Management System. Retrieved from https://continusys.com/7-benefits-of-an-integrated-business-management-system CPU Research and Development Center. (2025). Design and Implementation of a Unified Sales and Inventory System for Multi-Unit Operations. Central Philippine University Research Journal. Cyborg ERP. (2026). Web Based University Management System Smart ERP for Universities. Retrieved from https://www.cyborgerp.com/web-based-university-management-system/



Department of Computer Studies

BSIT Capstone Project Proposal Outline

	Department of Information and Communications Technology (DICT). (2024). Efforts on Digitalization of Frontline Services in the Philippine Government Setting. DevRova. (2025). Integrated Management Solutions: Transforming Business Efficiency. Retrieved from https://devrova.com/articles/integrated-management-solutions-business-efficiency/ Finale Inventory. (2025). Accounting and Inventory Software: Complete Guide to Integrated Business Management. Retrieved from https://www.finaleinventory.com/accounting-and-inventory-software Humedit Philippines. (2025). The Future of Business System is Integration. Retrieved from https://humedit.ph/accounting/the-future-of-business-system-is-integration/ Hudutech Ventures. (2025). ERP Systems for Educational Institutions: Centralized Management of Revenue-Generating Units. Hudutech Research and Publications. IJCRT. (2025). Web Based System For Seamless College Management. International Journal of Creative Research Thoughts. International Burch University. (2025). Inventory System for Small Businesses. Retrieved from https://eprints.ibu.edu.ba/ Juleb. (2025). Scale Smarter with an Integrated Accounting System. Retrieved from https://juleb.com/blog/Scale-Smarter-with-an-Integrated-Accounting-System/en Method CRM. (2025). Managing Multiple Business Operations Through a Unified Platform. Method CRM Official Blog. Mendoza, L., Santos, A., & Reyes, C. (2024). Technology and Accountability: Leveraging Integrated Systems for COA Compliance in State Universities. Asia Pacific Journal of Multidisciplinary Research. Mokogwu, et al. (2024). Data security and compliance in public sector systems. Journal of Digital Governance. National Treasury. (2026). IFMIS: Integrated Financial Management Information System. Retrieved from https://www.treasury.go.ke/ifmis/
--	--



Department of Computer Studies

BSIT Capstone Project Proposal Outline

	On Wave Group. (2024). How Integrated Business Systems Improve Efficiency. Retrieved from https://onwavegroup.com/blog/integrated-business-systems Process Point. (2025). Overcoming Information Silos: The Role of Centralized Systems in Multi-Department Organizations. Process Quodala, F. (2022). An Assessment of the Digitalization at the College and University Level. RSU Research Conference Proceedings. Rajan, S., & Sulthana, A. (2022). Modular System Design and Scalable Architecture for Business Applications. International Journal of Computer Science and Network Security, 22(4), 112–119. Republic Act No. 11032. (2025). E-Governance Act of the Philippines. Semantic Scholar. (2024). Cloud-Based Database Management and Secure Access Control in Integrated Business Systems. Journal of Information Technology and Application. The file is too long. System. Retrieved from https://juleb.com/blog/Scale-Smarter-with-an-Integrated-Accounting-System/en ClickBalance. (2025). La integración entre gestión comercial, inventario y finanzas: clave para la rentabilidad de tu PyME. The Future of Business System is Integration. Retrieved from https://humedit.ph/accounting/the-future-of-business-system-is-integration
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PROJECT WORKPLAN (system developer plan)

PROJECT OBJECTIVES	TARGET ACTIVITIES	OBJECTIVELY VERIFIABLE INDICATORS/TARGET ACCOMPLISHMENTS	Month								
			1	2	3	4	5	6	7	8	9
Objective 1 : Objective 1: Develop the Online Platform for Sales Transaction Management	- Conduct requirements gathering and stakeholder interviews - Design wireframes for sales transaction interface - Build frontend HTML/CSS/JS pages for sales recording - Implement Firebase Firestore CRUD for sales data - Integrate Node.js API for transaction processing	Fully functional sales transaction module with real-time Firestore recording									
Objective 2: Implement the Inventory Monitoring System	- Design database schema for inventory in Firestore - Develop product listing and stock tracking pages - Implement automated low-stock alert via Firebase Functions - Build restock notification system	Real-time inventory monitoring with automated stock alerts operational									



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PROJECT OBJECTIVES	TARGET ACTIVITIES	OBJECTIVELY VERIFIABLE INDICATORS/TARGET ACCOMPLISHMENTS	Month								
			1	2	3	4	5	6	7	8	9
Objective 3: Create the Financial Reporting Module	- Design report templates (daily, monthly, revenue) - Develop KPI dashboard with dynamic charts - Implement ROI calculation features - Enable printable report generation	Financial reports auto-generated and printable; KPI dashboard live									
Objective 4: Centralize Data Management	- Integrate all modules into unified Firestore database - Implement system logs and audit trail - Set up automated database backup routines	All data centralized in Firestore; audit logs and backups confirmed									
Objective 5: Improve Workflow Efficiency	- Integrate sales, inventory, and finance modules end-to-end - Eliminate redundant data entry through automation - Conduct internal workflow testing with OBA staff	Unified cross-module workflow operational; manual data entry reduced									
Objective 6:	Develop auto-computation logic for sales	All computations automated and verified against									



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PROJECT OBJECTIVES	TARGET ACTIVITIES	OBJECTIVELY VERIFIABLE INDICATORS/TARGET ACCOMPLISHMENTS	Month								
			1	2	3	4	5	6	7	8	9
Implement Automatic Computation of Financial Transactions	totals, change, and balances Implement daily/weekly/monthly revenue aggregation	manual calculations									
User Acceptance Testing & System Evaluation	- Deploy system to test environment - Conduct UAT with OBA administrators and staff - Collect ISO 25010 evaluation feedback - Fix reported bugs and finalize system	UAT sign-off obtained; evaluation results documented and analyzed									
Deployment, Training & Documentation	- Deploy system to live server / Firebase Hosting - Conduct staff training sessions - Prepare and submit user manual and final documentation	Live system deployed; final report and manual submitted									